

Exercise 5.1

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
In [2]: df = pd.read_csv('../data/velocities.txt', sep='\t', names=['t', 'v'])
df
```

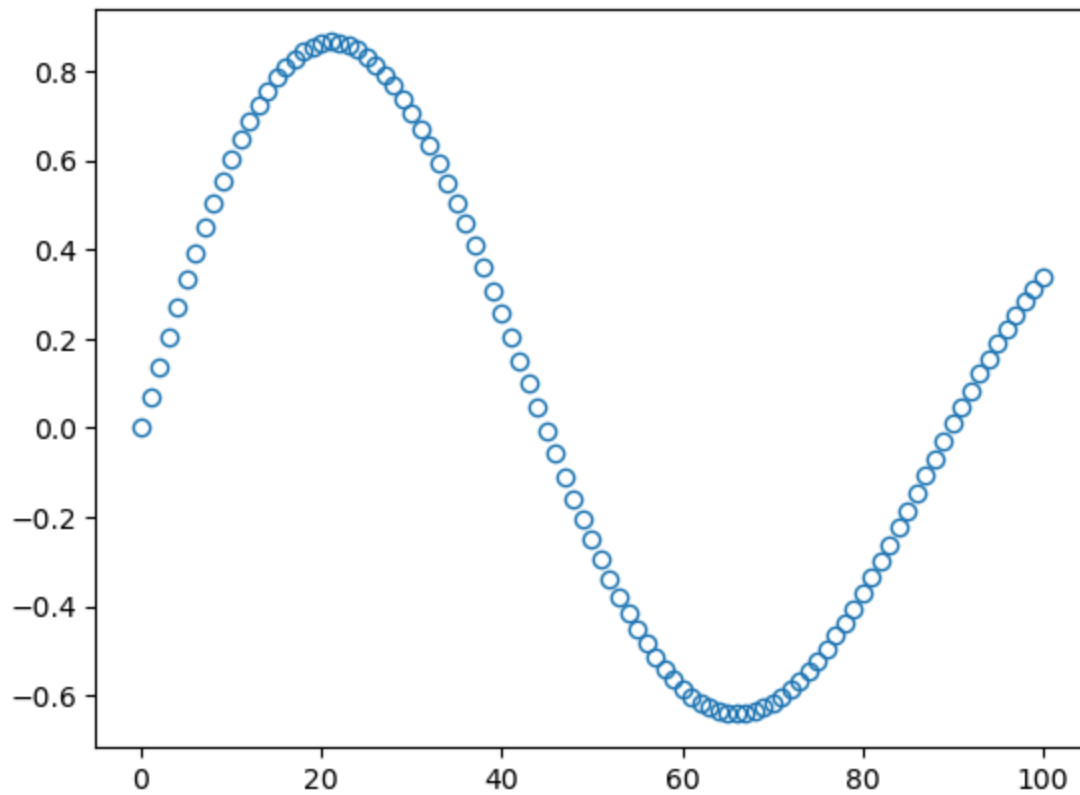
```
Out[2]:
```

	t	v
0	0	0.000000
1	1	0.069478
2	2	0.137694
3	3	0.204332
4	4	0.269083
...	...	...
96	96	0.223073
97	97	0.254244
98	98	0.283753
99	99	0.311479
100	100	0.337308

101 rows × 2 columns

```
In [4]: fig, ax = plt.subplots()
ax.plot(df['t'], df['v'], 'o', mfc='none')
```

```
Out[4]: [<matplotlib.lines.Line2D at 0x79c07d263d70>]
```



```
In [ ]: def trap(data):  
    dx = 1  
    x = data['t']  
    y = data['v']  
    s = dx/2*(func(a)+func(b))  
    s = s + np.sum(y[1:steps]*dx)  
    return(s)
```